

Kidney Disease in Liver, Cardiac, Lung, and Hematopoietic Stem Cell Transplantation

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Kidney disease can complicate all forms of nonrenal solid organ transplantation (SOT) and is associated with longer hospitalization, higher morbidity and mortality, and greater expense. Although there are organ-specific factors that impact on the incidence and severity of kidney disease, some useful generalizations can be made.

GENERIC KIDNEY ISSUES IN NONRENAL SOLID ORGAN TRANSPLANTATION

Estimation of Glomerular Filtration Rate

Transplant candidates and recipients often have low muscle mass and low creatinine generation. Hence, a mildly elevated creatinine concentration may represent a markedly reduced glomerular filtration rate (GFR). Such patients should have their GFR estimated by 24-hour urinary creatinine clearance (which will tend to overestimate because of tubular secretion) or, preferably, measured directly by iothalamate or other radioisotopes. Of the various creatinine-based equations that estimate GFR, the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) and Modification of Diet in Renal Disease (MDRD) equations appear to correlate best with true posttransplantation GFR.¹

Calcineurin Inhibitor Nephrotoxicity

Calcineurin inhibitors (CNIs; e.g., cyclosporine, tacrolimus) are the cornerstone of immunosuppression in SOT recipients, but also directly contribute to acute kidney injury (AKI) and CKD. CNIs also exacerbate posttransplantation hypertension and diabetes, both of which can cause CKD in the long term. It is probable that CNI nephrotoxicity is more severe in nonrenal (than kidney) transplantation because higher blood concentrations of CNIs may be used; furthermore, the denervated transplanted kidney may be protected from early sympathetic-mediated injury.²

Acute CNI nephrotoxicity is mainly a prerenal syndrome caused by reversible vasoconstriction of the afferent glomerular arteriole; tubular damage and microvascular disease may occur in severe cases. Acute CNI-induced thrombotic microangiopathy (TMA) is rare after SOT but is associated with a poor prognosis. In many cases TMA is precipitated by excessively elevated blood levels of CNI, or by infection, allograft rejection, or addition of a mammalian target of rapamycin (mTOR) inhibitor to CNI-based immunosuppression. Treatment usually involves discontinuation of the CNI (and/or mTOR inhibitor).

Electrolyte and other metabolic disturbances may be seen with acute and/or chronic CNI toxicity. Hyperkalemia, hyperuricemia, metabolic acidosis, and hypomagnesemia have been described, due to effects on renal tubular function.

Chronic CNI nephrotoxicity is probably the result of prolonged kidney ischemia and other effects, such as direct stimulation of kidney fibrogenesis. The patient is typically hypertensive with a steady fall in GFR, most marked in the first 6 to 12 months after transplantation

(Fig. 69.1).^{2,3} Urinalysis usually shows minimal/no hematuria and minimal/mild proteinuria. In practice, biopsies are often reserved for those with clinical features suggestive of a kidney disorder other than CNI toxicity.

Histologic examination shows changes in all compartments with “striped” interstitial fibrosis and tubular atrophy, arteriolar hyaline sclerosis, arteriosclerosis, and secondary focal glomerulosclerosis.² There may also be features of chronic TMA.² However, these “typical” histologic findings of CNI toxicity are nonspecific, making it difficult to estimate the actual contribution of CNI toxicity to post-transplant CKD.

The presumed high prevalence of chronic CNI nephrotoxicity has generated interest in low- or zero-dose CNI protocols (discussed later).² There remains reluctance to pursue these protocols because CNIs are such effective immunosuppressants, and organ replacement therapy analogous to dialysis is not available for other SOT recipients if severe rejection occurs. Another strategy is to preferentially use tacrolimus over cyclosporine, given that tacrolimus provides equivalent or superior immunosuppression with less nephrotoxicity.⁴ The roles of calcium channel blockers, renin-angiotensin system (RAS) blockers, and spironolactone in ameliorating CNI nephrotoxicity remain controversial. Control of hypertension is probably more important than the use of a particular antihypertensive agent.

Pretransplant Acute Kidney Injury

AKI is common in the days to weeks immediately before liver, heart, or (less commonly) lung transplantation and is typically prerenal (e.g., kidney hypoperfusion or hepatorenal syndrome), intrarenal (ischemic or toxic tubular injury), or a combination. Management focuses on treatment of the underlying cause and provision of dialytic support according to standard criteria. Because the patients are critically ill, continuous renal replacement therapy (CRRT) may be preferred to intermittent hemodialysis (HD), but there are no randomized controlled trials (RCTs) showing better outcomes with CRRT.

When severe or prolonged AKI occurs before anticipated transplantation, estimation of reversibility becomes very important, because presumed irreversible severe AKI shifts the management toward simultaneous dual-organ (e.g., liver plus kidney) transplantation. Kidney biopsy may help prognosticate but is not commonly performed because of technical difficulties in critically ill patients, who often also have coagulopathy.

The advantages and disadvantages of dual-organ transplantation are shown in Box 69.1. There is uncertainty about whether dual organ transplantation represents the best use of scarce donor organs, and some suggest prioritizing kidney transplantation after SOT instead. Liver, lung, and heart allocation is based on medical urgency, whereas kidney allocation is generally based on waiting time and immunologic factors. As a result, multiorgan transplants assume the priority of the

Changes in Measured Iothalamate GFR in Individual Patients After Lung Transplantation

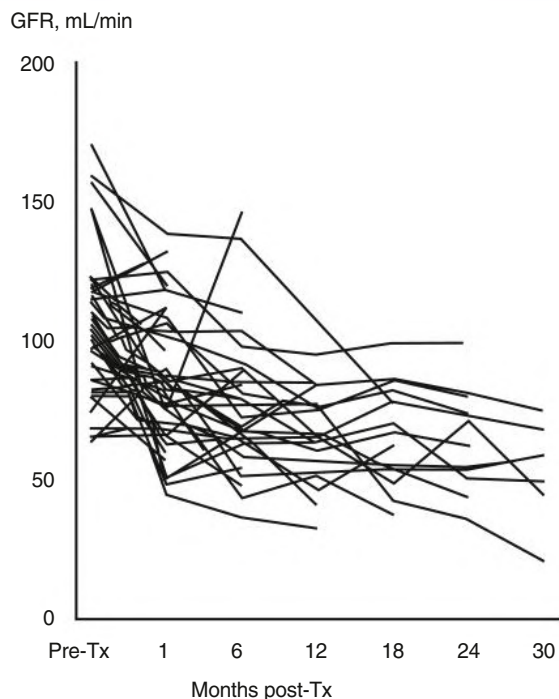


Fig. 69.1 Changes in Measured Iothalamate Glomerular Filtration Rate (GFR) in Individual Patients After Lung Transplantation. Note the variation in GFR before transplant and the large fall in GFR in many patients within the first 6 months. Tx, Transplantation. (Modified from Navis G, Broekroelofs J, Mannes GP, et al. Renal hemodynamics after lung transplantation. A prospective study. *Transplantation*. 1996;61[11]:1600–1605.)

BOX 69.1 Advantages and Disadvantages of Simultaneous Kidney and Nonrenal Transplantation

Advantages

- Potentially provides much better kidney function over the short and long term
- Single donor: potential for lower cumulative dose of immunosuppression (as opposed to kidney transplantation later)

Disadvantages

- Surgery is more technically complex and prolonged
- Deprives patients with “definite” end-stage kidney disease of a kidney transplant
- Not needed when the acute kidney injury is reversible

nonkidney organ and may therefore divert kidneys from those with “definite” end-stage kidney disease (ESKD), who may wait a long time for a kidney-only transplant. The kidney in a multiorgan transplant typically has a low kidney donor profile index (KDPI), again diverting these kidneys from the kidney-only wait list, where they would have likely been offered to pediatric recipients or other priority groups.⁵ Recent guidelines and policy updates have tried to address some of these controversies and are discussed later.^{5,6}

BOX 69.2 Causes of Acute Kidney Injury After Solid Organ Transplantation

Prerenal

- Hypovolemic shock (e.g., aggressive diuresis)
- Cardiogenic shock (e.g., severe cardiac allograft dysfunction)
- Distributive shock (e.g., sepsis)
- Cyclosporine or tacrolimus

Intrarenal (Acute Tubular Injury)

- Prolonged shock
- Cyclosporine or tacrolimus
- Aminoglycosides, amphotericin
- Intravenous contrast
- Massive hemolysis

Postrenal

- Rare

Posttransplant Acute Kidney Injury

AKI is common in the days to weeks after transplant and is often multifactorial (Box 69.2). It is associated with higher mortality and increased risk for CKD in survivors.⁷ Delayed introduction or reduced doses of CNIs may be used, usually under the cover of induction antibody immunosuppression.⁷

Passenger lymphocyte syndrome is uncommon but can occur in the early weeks after transplant (usually liver, but also other SOT) in the setting of a minor ABO or other blood group system mismatch. Donor lymphocytes (“passengers” with the transplant) produce anti-blood group antibodies, which cause hemolysis. Severe cases may be complicated by AKI.

Severe rhabdomyolysis has been reported, typically when a statin is prescribed with cyclosporine and an inhibitor of the cytochrome P-450 system, such as diltiazem or an azole-based antifungal agent.

BK polyomavirus nephropathy should be considered in the differential diagnosis of unexplained kidney dysfunction after SOT, especially in the context of intensive immunosuppression. However, the absolute risks of BK viremia and nephropathy remain low, and routine screening is not recommended.⁸

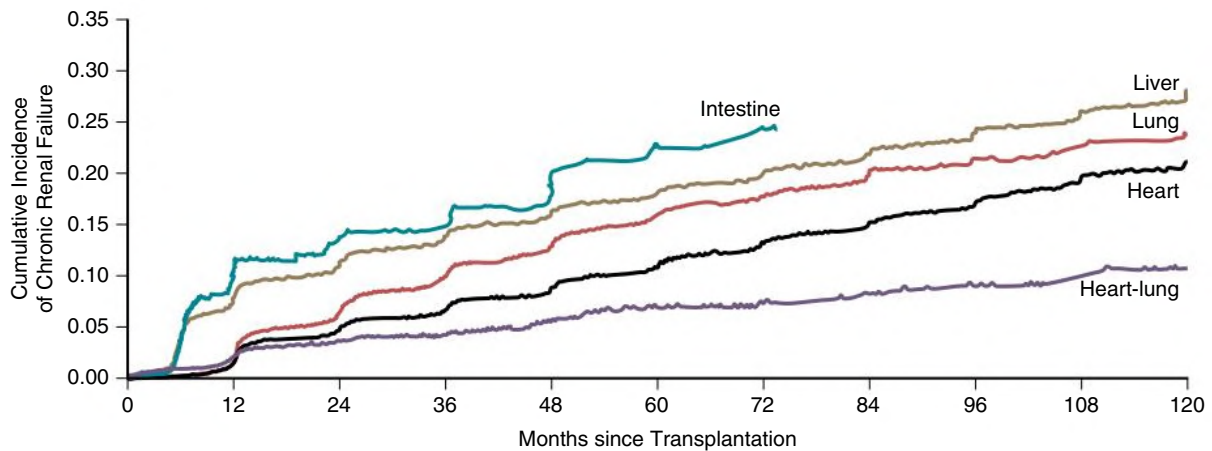
Chronic Kidney Disease

With the growing number and longer survival of transplant recipients worldwide, the absolute number of recipients with CKD has increased. The CKD-EPI equation (despite its limitations) is useful in identifying CKD in this setting.

In the most comprehensive study to date, the cumulative incidence of posttransplant CKD (defined as estimated GFR < 30 mL/min/1.73 m²) at 5 years varied according to organ transplanted, from 6.9% (heart-lung) to 21.3% (intestinal) (Fig. 69.2).⁷ Pretransplant variables that increase the risk for CKD include older age, female sex, lower GFR, diabetes, hypertension, hepatitis C virus, and need for dialysis. Posttransplant variables were also implicated: postoperative AKI and initial use of cyclosporine (as opposed to tacrolimus). Minimizing posttransplantation AKI and controlling hypertension and diabetes are therefore likely to prevent or slow CKD. Many studies have confirmed that CKD (particularly ESKD) after SOT portends a poor prognosis (Fig. 69.3), as well as increased hospitalization and infection rates.

Referral to a nephrologist should be considered in those with progressive decline in eGFR. The initial evaluation should include a thorough review of kidney function pre- and peritransplant, as well as exposure to nephrotoxins. If urinalysis shows moderate/severe

Cumulative Incidence of Stage 4 or 5 Chronic Kidney Disease in Nonrenal Organ Transplant Recipients



No. at Risk											
Heart-lung	576	375	295	219	194	156	133	107	72	46	30
Heart	24,024	19,885	17,238	14,687	12,341	10,022	7997	6104	4526	3096	1991
Intestine	228	152	110	84	57	33	23	13	8	5	5
Liver	36,849	28,495	24,041	19,508	15,724	12,564	9844	7345	5292	3614	2261
Lung	7643	5633	4316	3184	2327	1629	1136	745	468	258	133

Fig. 69.2 Cumulative incidence of chronic kidney disease (CKD) (estimated GFR <30 mL/min/1.73 m²) after transplantation of various solid organs. (Modified from Ojo AO, Held PJ, Port FK, et al. Chronic renal failure after transplantation of a nonrenal organ. *N Engl J Med*. 2003;349[10]:931–940.)

Mortality Associated With Chronic Kidney Disease in Organ Transplant Recipients

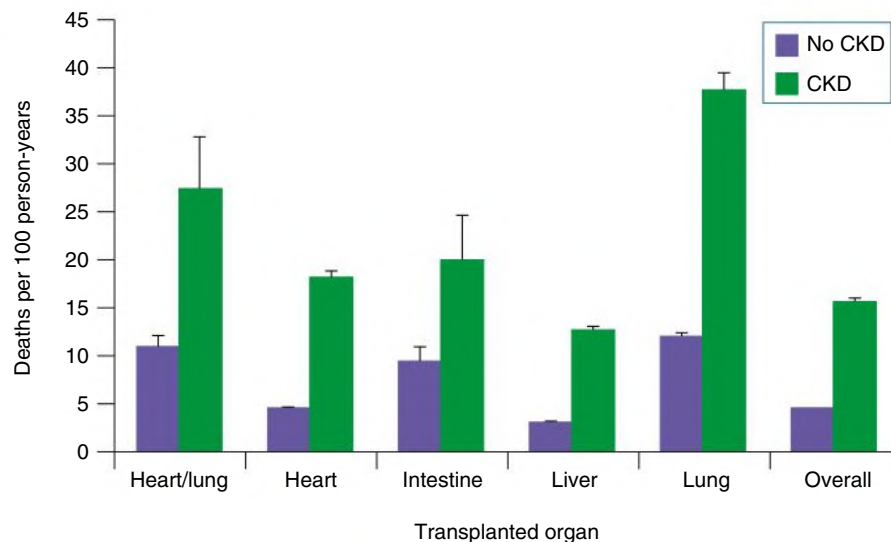


Fig. 69.3 Mortality associated with stage 4 and 5 chronic kidney disease (CKD) in organ transplant recipients. (From Scientific Registry of Transplant Recipients.)

proteinuria or hematuria or both, biopsy should be considered, as it may identify pathologies other than CNI nephrotoxicity.

Close consultation with the primary transplant team is important. When CNI toxicity is deemed the main cause of CKD, several approaches can be taken to reduce ongoing CNI toxicity and CKD progression.

A reduction in CNI dosage with a compensatory increase in anti-metabolite dosing for patients at low immunologic risk should be considered. If the patient is on a cyclosporine-based regimen, a switch

from cyclosporine to tacrolimus may be beneficial. An alternative strategy (one that we favor less) is to substitute an mTOR inhibitor for the CNI. This approach is not recommended in the early posttransplant period because of the impact of mTOR inhibitors on wound/anastomosis healing. The mTOR inhibitors themselves have some important kidney adverse effects: experimental models demonstrate delayed recovery from AKI due to effects on tubular cell proliferation. The mTOR inhibitors also potentiate the nephrotoxicity of cyclosporine when coadministered and can induce or exacerbate proteinuria.

CKD complications may be exacerbated by other posttransplant factors. For example, antiproliferative immunosuppressants may exacerbate anemia, and corticosteroids may exacerbate CKD-related mineral and bone disorder (CKD-MBD). In the absence of specific studies, it seems reasonable to apply standard guidelines to manage these and other CKD complications. Although there is experimental evidence that RAS blockers may have antifibrotic effects in nonrenal transplant CKD, there are no trials showing improved kidney outcomes with these compared with other antihypertensive agents.

End-Stage Kidney Disease

There is little doubt that kidney transplantation is the best form of RRT in SOT recipients with advanced CKD who are fit enough for the procedure. Where feasible, a preemptive living donor transplant is the best option. Maintenance immunosuppression is typically increased in the perioperative period; induction immunosuppression is not without risk and is not used in all cases.⁹ In the immediate posttransplant period, mortality is higher than remaining on the waiting list (reflecting the complications of surgery and more immunosuppression), but mortality in the medium to long term is much lower.⁷ The number of patients with a nonrenal transplant subsequently wait-listed for a kidney transplant in the United States has increased dramatically over the last 15 years.¹⁰ Mortality is much higher in SOT recipients wait-listed for kidney transplant than in wait-listed controls.¹⁰

Patients who do not receive a preemptive kidney transplant start dialysis for the usual indications. Home dialysis (home hemodialysis and peritoneal dialysis) should be explored and modality choice individualized for each patient.¹¹

KIDNEY DISEASE IN LIVER TRANSPLANTATION

AKI in cirrhosis has been defined as an absolute rise in serum creatinine of 0.3 mg/dL or greater ($\geq 26.5 \mu\text{mol/L}$) within 48 hours or a 50% or greater increase from baseline over 7 days.¹² AKI is common before liver transplantation and is usually due to hepatorenal syndrome (HRS), acute tubular necrosis (ATN), or both. Prolonged HRS may cause ATN.

If RRT is required in patients with severe HRS, CRRT is often preferred, in part because it may have less effect on intracranial pressure than intermittent dialysis. The need for dialysis in liver transplant candidates portends a very poor prognosis if liver transplantation does not occur.¹³ Some centers use intraoperative CRRT during liver transplantation in patients with severe preoperative AKI who are predicted to tolerate the surgery poorly from a volume or electrolyte perspective. This practice remains somewhat controversial.¹⁴

The Model for End-Stage Liver Disease (MELD) scoring system measures severity of liver dysfunction based on international normalized ratio, serum creatinine and bilirubin, and predicts 90-day mortality risk. The MELD-Na score includes sodium to further refine the short-term prognosis and has become the tool of choice to guide allocation of deceased donor liver allografts in many jurisdictions. As both tools include serum creatinine, patients with severe AKI are prioritized.

Posttransplantation AKI occurs in over 60% of recipients of donation after circulatory death (DCD) grafts and approximately 40% of recipients of donation after brain death (DBD) grafts or living donor grafts.¹⁵ Requirement for RRT is associated with higher mortality rates.

Although relatively low doses of CNIs are traditionally used in liver transplantation (because rejection is less of a concern than in other SOT), CKD remains common, suggesting other factors (such

as diabetes and hepatitis virus-associated glomerulonephritis) play an important role.

Various protocols involving reduction or cessation of CNIs have been studied with mixed results. Induction with interleukin-2 (IL-2) receptor blockers, and maintenance with mycophenolate mofetil (MMF), corticosteroid, and delayed (day 5) introduction of low-dose tacrolimus was less nephrotoxic than other regimens while maintaining excellent allograft and patient outcomes.¹⁶ Conversion to sirolimus does not have clearly proven benefits and is associated with certain risks, and so is not routinely performed. CNI withdrawal to MMF monotherapy was shown to be a reasonable option for selected patients who were several years out from transplant.¹⁷

The number and percentage of simultaneous liver-kidney (SLK) transplants in the United States have increased since the MELD score was adopted into allocation policy, such that over 700 SLK transplants are performed annually ($>8\%$ of all liver transplants).⁶ This is much higher than in many other jurisdictions (e.g., Canada, United Kingdom), where SLK accounts for fewer than 2% of all liver transplants.

Although selected patients with severe pretransplant kidney disease benefit from SLK transplant, it can be difficult to predict which patients are likely to have meaningful kidney recovery without kidney transplant. Prognostication is further complicated by the risk of kidney biopsy in this patient cohort.

In the United States the Organ Procurement and Transplantation Network (OPTN) has provided updated clinical recommendations (which do not mandate a kidney biopsy) as to who should receive SLK transplants (Box 69.3).⁶ This policy also includes a safety net of priority for kidney transplant in those patients who have advanced CKD following liver transplantation ($\text{eGFR} \leq 20 \text{ mL/min/1.73 m}^2$ or dialysis dependent and registered on the kidney transplant wait list in the first year post-liver transplant). The Canadian Forum guidelines suggest that patients are eligible for SLK if they are dialysis dependent for more than 3 months. They are “possibly eligible” (determined on a case-by-case basis) if they are on dialysis for a shorter time or if they have an eGFR less than $30 \text{ mL/min/1.73 m}^2$ for more than 1 month.⁵

BOX 69.3 UNOS and OPTN Policy Regarding Candidacy for Simultaneous Liver-Kidney Transplant in Patients Listed for Liver Transplant

- CKD defined as $\text{GFR} \leq 60 \text{ mL/min/1.73 m}^2$ for >90 consecutive days and one of the following:
 - Dialysis dependence (ESKD)
 - $\text{GFR} \leq 30 \text{ mL/min/1.73 m}^2$ at transplant wait-listing
- AKI defined as ≥ 6 weeks of one or a combination of the following:
 - $\text{GFR} \leq 25 \text{ mL/min/1.73 m}^2$ at least once every 7 days
 - Dialysis at least once every 7 days
- Specific metabolic diseases
 - Hyperoxaluria
 - Atypical hemolytic uremic syndrome (from factor H or I deficiency)
 - Methylmalonic aciduria
 - Familial nonneuropathic systemic amyloidosis

AKI, Acute kidney injury; CKD, chronic kidney disease; ESKD, end-stage kidney disease; GFR, glomerular filtration rate; OPTN, Organ Procurement and Transplantation Network; UNOS, United Network for Organ Sharing.

Modified from Boyle G. Simultaneous Liver Kidney (SLK) Allocation Policy. https://optn.transplant.hrsa.gov/media/1192/0815-12_SLK_Allocation.pdf.

KIDNEY DISEASE IN CARDIAC TRANSPLANTATION

Preoperative AKI is common, and approximately 1% of heart transplant recipients require RRT before transplantation.⁷ The main cause of AKI is kidney hypoperfusion from severe congestive heart failure (type 1 cardiorenal syndrome). In some patients there may also be a component of CKD for a variety of reasons, including chronic hypoxia, renovascular disease, hypertension, and atheroembolism. Ventricular assist devices are increasingly used as a bridge to cardiac transplantation and may improve kidney perfusion and function in this setting.

Postoperative AKI is also common, with 5% to 30% of cardiac transplant recipients requiring dialysis, depending on patient- and center-specific characteristics.¹⁸ There is a strong association between the severity of postoperative AKI and both mortality and kidney dysfunction at 1 year.¹⁸ In addition to the causes shown in Box 69.2, risk factors for AKI after cardiac transplant surgery are prolonged aortic cross-clamping, large fluid volume shifts, prolonged cardiac allograft ventricular dysfunction, postoperative bleeding, and reexploration.

CKD is an important complication in the medium to long term and portends a worse prognosis¹⁹ (see Fig. 69.2). The cumulative incidence of stage 4 or stage 5 CKD in cardiac transplant recipients has been reported as 11% at 5 years, which is lower than in bowel, liver, or lung transplant recipients.⁷

The principal cause of CKD after cardiac transplantation is CNI toxicity. A number of studies have assessed the impact of CNI modification, reduction, or withdrawal after cardiac transplantation. A small RCT compared tacrolimus to cyclosporine and reported a favorable side effect profile (including kidney function) with tacrolimus, with similar survival and rejection rates.²⁰ Since this study, tacrolimus use has continued to increase (and the use of cyclosporine greatly decreased), such that more than 95% of cardiac transplant recipients now receive tacrolimus-based immunosuppression.²¹ Despite some encouraging short- and medium-term results with everolimus and reduced CNI dosing in patients at low immunologic risk, fewer than 10% of cardiac transplant recipients receive an immunosuppression regimen that includes everolimus or sirolimus.²²

The absolute number of simultaneous heart-kidney transplants remains low but is steadily increasing in the United States (in 2018 approximately 7% of total heart transplants).²³ The criteria for simultaneous heart-kidney transplant eligibility are generally less well-defined than in the case of SLK, and the SLK criteria (Box 69.3) are often used as a reference point.⁸

KIDNEY DISEASE IN LUNG TRANSPLANTATION

Preoperative AKI is less common, with 0.1% of lung transplant recipients requiring dialysis before transplantation.⁷ Postoperative AKI is common and associated with significantly reduced survival. A systematic review and meta-analysis showed that approximately 53% of patients had postoperative AKI (KDIGO criteria), and 9.3% required RRT after lung transplantation.²⁴

AKI can occur for the usual reasons (see Box 69.2), but factors specific to lung transplantation include aggressive diuresis and nephrotoxic antimicrobial agents for resistant infections (e.g., aminoglycosides, amphotericin). Acute oxalate nephropathy leading to irreversible kidney failure has been described. It is thought that enteric hyperoxaluria is caused by either prolonged antibiotic administration (which interferes with colonic flora) and/or cystic fibrosis-related pancreatic exocrine insufficiency.²⁵ Although probably rare, this “new disease” emphasizes the importance of maintaining a broad differential diagnosis for kidney disease occurring after SOT and the importance of biopsy when the clinical manifestation is unusual.

CKD, although often underappreciated, is common in the first year after lung transplantation, and the incidence increases thereafter. As with other forms of posttransplantation CKD, the most common cause is probably CNI toxicity. CNI dosages tend to be higher because of the increased risk for rejection in lung transplantation. Although CNI withdrawal is poorly tolerated in this setting, a few studies have reported improvements in GFR when doses of CNIs were reduced. CNI reduction should be accompanied by addition/optimization of MMF. De novo use of mTOR inhibitors is not recommended because they may be associated with breakdown of the bronchial anastomosis; later use is associated with pneumonitis. Simultaneous lung-kidney transplantation is very rarely performed.

KIDNEY DISEASE IN HEMATOPOIETIC STEM CELL TRANSPLANTATION

Hematopoietic stem cell transplantation (HSCT) involves the administration of chemotherapy with or without radiotherapy, followed by bone marrow engraftment of stem or progenitor cells.^{26,27} Most commonly, HSCT is used to treat hematologic malignancies; however, other indications include certain solid tumors, nonmalignant hematologic conditions (e.g., aplastic anemia, sickle cell disease), genetic immunodeficiencies, and inborn errors of metabolism.²⁸ Stem and progenitor cells may be harvested from peripheral blood, bone marrow, or umbilical cord blood, and may come from the affected patient (*autologous*) or a related or unrelated donor (*allogeneic*). In both *myeloablative* autologous and allogeneic HSCT, intensive pretransplant conditioning regimens using high-dose chemotherapy and radiation are given to eradicate the bone marrow and underlying cancer; the hematopoietic system is then reconstituted by infusion and engraftment of transplanted cells. Patients not eligible for aggressive myeloablative therapy because of older age or comorbidities may be candidates for *nonmyeloablative* (low-intensity) or *reduced-intensity conditioning* (RIC; intermediate intensity) regimens. These less toxic conditioning regimens provide sufficient immunosuppression to permit the engraftment of transplanted cells, which then target tumor cells (the graft-versus-tumor effect).²⁹ Both myeloablative and nonmyeloablative allogeneic transplantation (but not autologous transplantation) require posttransplant immunosuppression to prevent acute and chronic graft-versus-host disease (GVHD); CNIs are often prescribed for this purpose.

Acute Kidney Injury After Hematopoietic Stem Cell Transplantation

Epidemiology

AKI is a common complication of HSCT, with reported incidence and severity varying by the conditioning regimen (myeloablative > nonmyeloablative), type of transplant (allogeneic > autologous), and definition of AKI used (Table 69.1). It is most common (>70% of patients in some series) after myeloablative allogeneic HSCT, likely reflecting the longer period of profound immunosuppression (with associated risk of sepsis) and much greater risk of hepatic sinusoidal obstruction syndrome (SOS), previously called venoocclusive disease (VOD) of the liver, compared with nonmyeloablative allogeneic regimens.^{27,30} Autologous transplantation carries the lowest risk of AKI (approximately 20%); this has been attributed to more rapid engraftment, the absence of GVHD, and lack of CNI exposure.³¹ AKI after HSCT confers poor short- and long-term prognoses.

The incidence of dialysis-requiring AKI ranges from 0% (autologous) to 33% (myeloablative allogeneic), with approximately 5% of patients requiring RRT overall. The prognosis is very poor for those

TABLE 69.1 Three Types of Hematopoietic Stem Cell Transplantation and Their Associated Kidney Complications

	Myeloablative Allogeneic	Myeloablative Autologous	Nonmyeloablative Allogeneic
Diseases treated	Many leukemias, lymphomas, myelodysplastic syndromes, myeloproliferative neoplasms, multiple myeloma, marrow failure states (e.g., aplastic anemia), select inherited immunodeficiencies, metabolic disorders, and hemoglobinopathies	Multiple myeloma, lymphomas, germ cell tumors, neuroblastoma	As for myeloablative allogeneic type
Used in patients >60 years	Rarely	Sometimes	Often
Comorbidities permissible before HSCT	Minimal	Minimal	Some
Intensity of conditioning regimen	High	High	Low (intermediate for RIC regimens)
GVHD after HSCT	Common	None	Common
CNIs used routinely	Yes	No	Yes
Mortality	Moderate-high	Low	Moderate-high
Risk of relapse	Lower	Higher	Lower
Incidence of AKI ³⁶	Very common (19%–66%)	Common (12%–50%)	Common (29%–54%)
Causes of AKI	Hepatic SOS, shock syndromes, nephrotoxic drugs, CNIs, acute TMA	Shock syndromes, nephrotoxic drugs, occasionally hepatic SOS	Shock syndromes, nephrotoxic drugs, CNIs, acute TMA
Incidence of CKD	Common	Common	Mild forms probably common
Causes of CKD	Irreversible AKI, kidney TMA, CNIs, GVHD	Irreversible AKI, pretransplant CKD (e.g., kidney involvement by amyloidosis or multiple myeloma)	Pretransplant mild CKD, irreversible AKI, CNIs, GVHD

AKI, Acute kidney injury; CKD, chronic kidney disease; CNIs, calcineurin inhibitors; GVHD, graft-versus-host disease; HSCT, hematopoietic stem cell transplantation; RIC, reduced-intensity conditioning; SOS, sinusoidal obstruction syndrome; TMA, thrombotic microangiopathy.

requiring dialysis (early mortality >80%).^{27,31} Fortunately, rates of post-HSCT AKI are decreasing, likely due to multiple improvements in peritransplant care.

Etiology

It is helpful to consider the causes of AKI according to the period after HSCT (Box 69.4).²⁶ *Tumor lysis syndrome* is an immediate but rare complication after HSCT, as the cancer is often eliminated by the time of transplant. *Marrow infusion syndrome* occurs within 24 to 48 hours of stem/progenitor cell infusion when patients are exposed to products of red blood cell lysis (released during cryopreservation), as well as the cryoprotectant dimethyl sulfoxide (DMSO), leading to intratubular pigment cast formation and/or toxic ATN.³² This complication is now infrequent, likely related to advances in cell preservation.

Within the first few weeks of HSCT, when intense conditioning regimens have caused pancytopenia, mucositis, vomiting, and liver damage, recipients are at high risk of many forms of AKI. These include prerenal syndromes (e.g., hypovolemia, capillary leak syndrome, engraftment syndrome, hepatic SOS) and acute tubular injury from sepsis. Medication-induced nephrotoxicity may take the form of acute tubular injury (aminoglycosides, amphotericin B, intravenous iodinated contrast), crystal-induced tubular toxicity (acyclovir, sulfonamides), or acute interstitial nephritis (many antibiotics, allopurinol, proton pump inhibitors).^{31,33} CNIs have toxicities as outlined previously.

Obstructive uropathy is much less common but can be caused by severe hemorrhagic cystitis or fungal infection of the collecting system. Causes of hemorrhagic cystitis include high-dose cyclophosphamide and viral infections (e.g., adenovirus, BK virus). Adenovirus and BK nephropathies have also been described.³¹

In addition to the prerenal AKI that occurs in the context of gastrointestinal GVHD, moderate and severe acute and chronic GVHD are correlated with intrarenal AKI and CKD, attributed in part to excessive release of proinflammatory cytokines.³³ Animal models of GVHD have demonstrated infiltration of donor-type leukocytes into the kidneys, with associated cell-mediated injury to the kidney microvasculature and kidney tubules.³⁴

Capillary Leak Syndrome and Engraftment Syndrome

Capillary leak syndrome and engraftment syndrome are caused by endothelial cell injury and the release of proinflammatory cytokines and are characterized by fluid retention and fever. Both lead to AKI due to intravascular volume depletion, and possibly by a direct inflammatory effect.³³ Capillary leak syndrome presents within 2 weeks of HSCT and is seen most often in haploidentical transplants as a component of cytokine release syndrome (CRS). Engraftment syndrome presents similarly to capillary leak syndrome but occurs at the time of neutrophil regeneration, and is seen most commonly after autologous HSCT.³³

Hepatic Sinusoidal Obstructive Syndrome (Previously Venoocclusive Disease of the Liver)

Hepatic SOS is an important cause of severe AKI after myeloablative HSCT, particularly myeloablative allogeneic HSCT (see Table 69.1 and Box 69.4). The incidence ranges from 2% (by updated European Society for Blood and Marrow Transplantation criteria) to 30%.³⁰ Pathophysiology is thought to involve radio- and chemotherapy-induced damage to the endothelium of hepatic venules with subsequent venular thrombosis and sinusoidal and

BOX 69.4 Causes of Kidney Disease According to Time After Hematopoietic Stem Cell Transplantation

Immediate (Very Rare)

- Tumor lysis syndrome
- Marrow infusion syndrome

Early (AKI in First 3 Months)

- Prerenal
 - Hypovolemia (gastrointestinal GVHD, drug-induced nausea and vomiting, salt and water losses due to diuretics or other tubular injury)
 - Capillary leak syndrome, engraftment syndrome
 - Hepatic SOS (early)
 - Heart failure
 - CNI toxicity (early)
- Intrarenal
 - Ischemic acute tubular injury (prolonged prerenal ischemia, sepsis/septic shock, hepatic SOS [late], calcineurin inhibitor toxicity [late], intravenous iodinated contrast)
 - Toxic acute tubular injury (aminoglycosides, amphotericin B, foscarnet, cidofovir)
 - Crystal-induced tubular toxicity (acyclovir)
 - Allergic interstitial nephritis (penicillins, cephalosporins, fluoroquinolones, macrolides, trimethoprim-sulfamethoxazole, allopurinol, proton pump inhibitors)
- Postrenal
 - Hemorrhagic cystitis (cyclophosphamide, BK virus, adenovirus)

Late

- Thrombotic microangiopathy
- Calcineurin inhibitor toxicity
- Irreversible AKI
- Membranous nephropathy or other glomerular diseases
- Recurrence of original disease, which then affects the kidneys (e.g., amyloidosis, multiple myeloma)
- Retroperitoneal fibrosis (malignancy, radiotherapy)
- GVHD (?)

AKI, Acute kidney injury; CNIs, calcineurin inhibitors; GVHD, graft-versus-host disease; SOS, sinusoidal obstruction syndrome.

portal hypertension that in turn decreases kidney perfusion. Independent risk factors include the use of antithymocyte globulin, myeloablative conditioning regimens (total body irradiation based and busulfan based), busulfan exposure (pediatric population), 2 or more HSCTs, and increased tacrolimus trough blood concentrations (>5–10 ng/mL).³⁰ The use of gemtuzumab ozogamicin and inotuzumab ozogamicin, novel antibody-drug conjugates sometimes administered prior to HSCT, have also been implicated.^{30,31}

Clinically, SOS presents as a form of hepatorenal syndrome during the first 30 days after HSCT. There is frequently a precipitating factor such as sepsis. Initial symptoms and signs include weight gain, edema, and ascites, followed by right upper quadrant abdominal pain, jaundice, and abnormal liver function tests. Oliguria and rising creatinine follow.³³ In mild to moderate cases, patients may recover with supportive care alone. Severe SOS complicated by liver and kidney failure (and frequently respiratory failure) carries a mortality approaching 80%.³⁵ The differential diagnosis includes acute GVHD of the liver, sepsis, drug-induced cholestasis, gallstone disease, and hepatotoxic effects of parenteral nutrition.

The diagnosis of hepatic SOS is usually based on characteristic clinical and laboratory features, though liver biopsy is occasionally performed. Urine sodium concentration is low (<10 mEq/L). Urinalysis and urine sediment are typically bland; granular casts may be seen with progression to ATN. Kidney biopsy specimens have not revealed intrinsic kidney lesions, consistent with the understanding that SOS is most likely hemodynamic in pathophysiology.³⁶

Transplant-Associated Thrombotic Microangiopathy

Transplant-associated thrombotic microangiopathy (TMA) is associated with subclinical kidney disease, AKI, and (most commonly) CKD. Recent studies reveal an incidence of 2.3% to 30%, with widely varying definitions of transplant-associated TMA and study populations.³⁷ Transplant-associated TMA is characterized by endothelial cell injury leading to microangiopathic hemolytic anemia, thrombocytopenia, and end-organ damage; the kidneys are thought to be vulnerable to microvascular injury due to their extensive glomerular circulation.³⁶ Patients typically present 4 to 12 months after HSCT with slowly rising creatinine, although some patients demonstrate a more fulminant presentation with severe nephritic syndrome. Clues to the diagnosis include disproportionate hypertension and progressive anemia. Urinalysis shows variable proteinuria and hematuria. Careful review of prior test results may show evidence of an intermittent or persistent low-grade TMA, although the classic laboratory findings (e.g., elevated serum lactate dehydrogenase, decreased haptoglobin, schistocytes) may not all be present. Diagnostic criteria (and their limitations) are summarized elsewhere.³⁸ Histopathology reveals endothelial cell swelling, fibrin thrombi within capillary loops and arterioles, fragmented red blood cells, mesangiolysis, and thickened arteriolar vasculature. Chronic injury leads to glomerular capillary wall remodeling, double contour formation of the basement membrane, and tubular injury with interstitial fibrosis (Fig. 69.4).³⁷

It is unclear if transplant-associated TMA is a direct complication of HSCT or a consequence of endothelial injury precipitated by GVHD or other exposures.^{31,37} Acute GVHD (especially grades 2–4) is a major risk factor for transplant-associated TMA, with emerging data suggesting that transplant-associated TMA might be an “endothelial complication” of GVHD.^{36,39} Other significant risk factors include unrelated donor type and CNI use. Transplant-associated TMA has also been associated with advanced primary malignancy, nonmyeloablative transplant, high-dose busulfan administration, total body irradiation (TBI), mTOR inhibitor use, and infections (e.g., systemic infections, *Aspergillus* species, BK virus, cytomegalovirus, parvovirus B19, and adenovirus).³⁷ Recent studies demonstrate that complement activation and genetic susceptibility to complement dysregulation play major roles in the endothelial injury that drives transplant-associated TMA, with potential implications for therapy.^{37,40} (Fig. 69.5).

Prevention, Detection, and Management of Acute Kidney Injury After Hematopoietic Stem Cell Transplantation

General Considerations

Strict control of fluid balance is essential for the prevention of posttransplant AKI. Weight, blood pressure, intake, and output (urine and stool) should be monitored regularly in all HSCT patients.³¹ Nephrotoxic medications and iodinated contrast should be used judiciously. Infections and gastrointestinal GVHD should be treated promptly. If CNI trough concentrations are high, dose reduction should be considered. Promising developments that may lower the risk of AKI include the use of CNI-free (e.g., cyclophosphamide-based) regimens for GVHD prophylaxis and personalized drug dosing protocols.³¹

As previously discussed, serum creatinine has limitations as an AKI biomarker in this context. Novel AKI biomarkers (e.g., urine

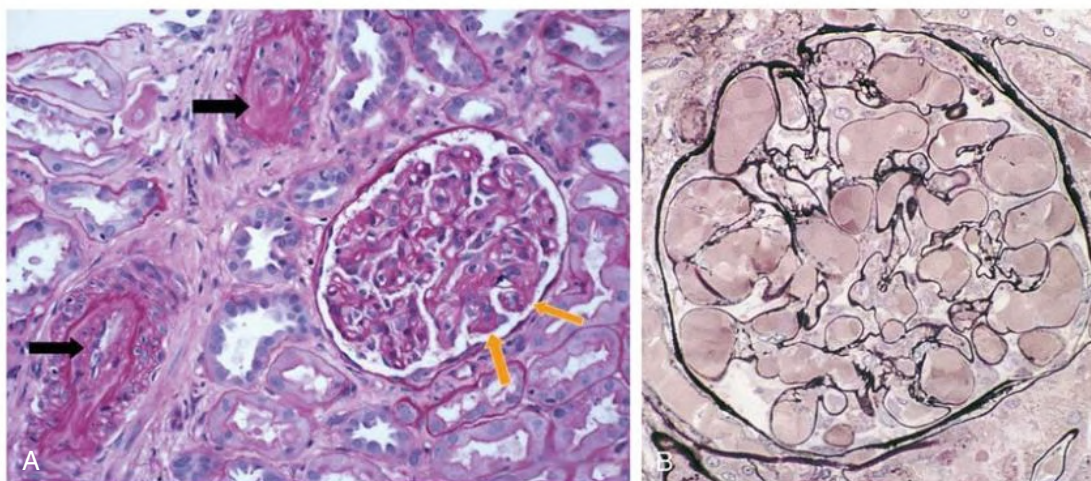


Fig. 69.4 Thrombotic Microangiopathy After Allogeneic Hematopoietic Stem Cell Transplantation (HSCT). (A) Kidney biopsy specimen from a patient who had undergone allogeneic HSCT and developed subacute kidney failure 12 months later. Periodic acid-Schiff staining shows near occlusion of two small arteries by subintimal connective tissue and swollen endothelium (*black arrows*). The glomerulus shows thickened capillary walls with “double contours” and segmental occlusion and collapse of capillaries (*orange arrows*). (B) Severe mesangiolysis in a patient with thrombotic microangiopathy after HSCT. Note the aneurysmal capillary loops and the lack of mesangial cells or matrix. (Courtesy Dr. H. Rennke, Harvard Medical School, Boston.)

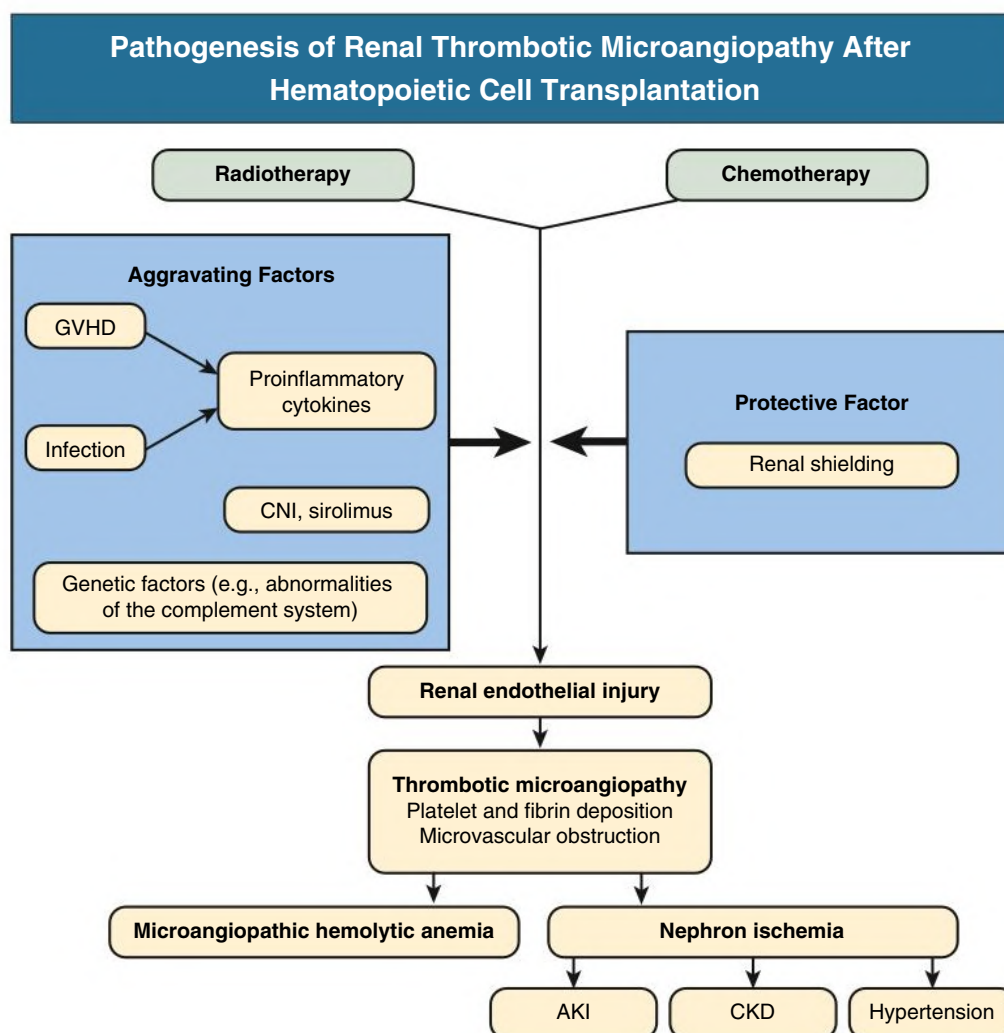


Fig. 69.5 Putative pathogenesis of kidney thrombotic microangiopathy after hematopoietic stem cell transplantation. AKI, Acute kidney injury; CKD, chronic kidney disease; CNI, calcineurin inhibitor; GVHD, graft-versus-host disease. (Modified from Humphreys BD, Soiffer RJ, Magee CC. Renal failure associated with cancer and its treatment: an update. *J Am Soc Nephrol*. 2005;16[1]:151–161.)

neutrophil gelatinase-associated lipocalin [NGAL]) have been evaluated in the research setting but are not yet in routine clinical use.³⁶

Posttransplant AKI should be evaluated as for any patient with hospital-acquired AKI but with particular focus on the patient's cancer diagnosis, type of transplant, and conditioning regimen. Complications specific to HSCT, such as capillary leak syndrome/engraftment syndrome, hepatic SOS, and transplant-associated TMA, should be considered in the appropriate clinical context. Nephrotoxic medications should be stopped where possible, and medications dose adjusted for eGFR.³¹ Urinalysis, urine microscopy, urine protein/creatinine ratio, and kidney ultrasound should be performed. Clinicians should have a low threshold to evaluate for TMA, BK virus, and adenovirus. Kidney biopsy may be considered in the following situations: (1) AKI of unclear etiology, (2) failure of kidney function to recover as anticipated after initial modifications in therapy, or (3) presence of nephrotic-range proteinuria. As many HSCT patients are thrombocytopenic and/or coagulopathic, biopsies should be performed with caution and appropriate transfusion support.³¹

The choice of RRT modality is usually based on hemodynamic stability and volume status; daily obligate fluid intake in this context is frequently massive, and fluid balance may be controlled most easily with continuous RRT. There is evidence in the pediatric literature that continuous convective therapies may be preferred, perhaps related to differences in the removal of small- and middle-sized molecules compared with diffusive therapies.⁴¹

Data supporting earlier (versus later) initiation of RRT after HSCT comes primarily from the pediatric population, with a small recent study demonstrating good outcomes in patients with SOS, AKI, and fluid overload treated with a standardized fluid balance protocol and early initiation of RRT.⁴² In the absence of consensus guidelines, some experts advocate for initiation of RRT when there is radiologic and clinical evidence of pulmonary edema, worsening oxygenation, poor response to high-dose diuretics, evidence of systemic inflammation, and ongoing oliguria.³⁶

Importantly, vascular access can be problematic in these patients because of thrombocytopenia and neutropenia, predisposing to bleeding and infection, respectively.

Disease-Specific Considerations

Hepatic sinusoidal obstruction syndrome. Current strategies for preventing hepatic SOS are based on inconclusive evidence and include avoiding precipitating factors where possible and using ursodeoxycholic acid (UDCA) or low-dose heparin.⁴³ Ursodeoxycholic acid alters the bile acid milieu, rendering bile less hydrophobic and toxic to parenchymal cells if cholestasis is present; it also exerts local cytokine-mediated immunomodulatory effects. Defibrotide, an oligonucleotide with antithrombotic and fibrinolytic effects on the microvascular endothelium but with apparently few systemic adverse effects, has demonstrated efficacy in both preventing and treating SOS.⁴⁴ As discussed earlier, continuous RRT may have less effect on intracranial pressure than intermittent HD.

Transplant-associated thrombotic microangiopathy. Patients with suspected or confirmed transplant-associated TMA should be supported with transfusions, blood pressure control, and dialysis as indicated.³⁷ All patients should be promptly evaluated for precipitating viral infections and acute GVHD. In patients receiving CNIs or mTOR inhibitors for GVHD prophylaxis, replacement with alternative agents (corticosteroids, MMF, IL-2 inhibitors, and anti-CD20 agents) may be appropriate, although not extensively studied.³⁸ Plasma exchange has not been proven to be beneficial.^{37,38} Limited reports in pediatric and adult patients have demonstrated response rates of 80% (12 of 15 patients) with the use of rituximab.³⁷ Given the complement pathway abnormalities described previously, there is interest in the C5 inhibitor eculizumab. Data for its use

comes primarily from the pediatric literature, with a relatively large study of 64 children with high-risk transplant-associated TMA demonstrating 66% 1-year post-HSCT survival compared with 16.7% in a previously reported untreated cohort.⁴⁵ Of course, the risks and benefits of any extra immunosuppression in patients after HSCT must be very carefully considered.

Chronic Kidney Disease After Hematopoietic Stem Cell Transplantation

Epidemiology

CKD is an important long-term complication of HSCT, particularly allogeneic HSCT.²⁷ The reported cumulative incidence of CKD after HSCT ranges from 7% to 48%, with approximately 4% of these patients progressing to ESKD and experiencing worse survival on dialysis compared with non-HSCT controls.^{27,38,46} The incidence of CKD after HSCT is expected to rise as the age of patients undergoing transplantation increases.³¹ Some causes of CKD are shown in Table 69.1.

Risk factors for posttransplant CKD include age 45 years or older at time of transplantation, baseline GFR less than 90 mL/min/1.73 m², acute and chronic GVHD, prior AKI, and survival more than 1 year after transplantation.²⁷ Exposure to high-dose total body irradiation has also been implicated, possibly by contributing to subacute or chronic transplant-associated TMA.^{28,31} Hypertension is a well-recognized late complication of HSCT, with incidence linked to the development of CKD. Albuminuria has been associated with CKD progression and decreased posttransplant survival.³¹

CNIs are routinely prescribed after allogeneic HSCT to prevent and treat GVHD. They are often stopped after 3 to 6 months (unless there is ongoing GVHD), and so their contribution to CKD is thought to be limited.²⁷ The contribution of CNIs to chronic kidney TMA is unclear, as transplant-associated TMA can certainly occur in patients on non-CNI protocols.⁴⁶

Glomerular Disease

Nephrotic syndrome has been described after both allogeneic (particularly nonmyeloablative) and autologous HSCT. Nephrotic syndrome after allogeneic HSCT tends to occur more than 6 months after transplantation and appears to be strongly associated with tapering of immunosuppression and the presence of GVHD; de novo membranous nephropathy is the most common biopsy finding, followed by minimal change disease.^{47,48} Importantly, the original hematologic disease (e.g., amyloidosis, multiple myeloma) may recur with kidney involvement.

Management of Hematopoietic Stem Cell Transplantation–Related Chronic Kidney Disease

Proteinuria and hypertension should be monitored and controlled. There is limited evidence regarding the superiority of RAS blockade, but its use seems reasonable.⁴⁹ Management of chronic transplant-associated TMA is discussed earlier. Clinical response in patients with post-HSCT membranous nephropathy or minimal change disease is usually seen with increased immunosuppression (typically corticosteroids and/or CNI); there are also limited reports of success with rituximab (particularly in membranous nephropathy).^{47,48}

Suitability for kidney transplantation should be judged on a case-by-case basis. On occasion, the allogeneic stem cell donor can donate a kidney; the benefit of this approach is that tolerance to the kidney allograft should exist; hence minimal or no additional immunosuppression is required.⁵⁰ If this option is not available and the patient receives a conventional kidney transplant, low-dose immunosuppression should generally be prescribed because HSCT recipients may not have normal immunity and remain at higher risk for infection. Good outcomes after kidney transplantation have been reported in carefully selected patients.^{50,51}

SELF-ASSESSMENT QUESTIONS

- A 28-year-old woman is referred to the nephrology clinic with serum creatinine 2.3 mg/dL (200 μ mol/L). Six years ago she underwent lung transplantation because of respiratory failure associated with cystic fibrosis. The lung transplant is working well. However, her serum creatinine has increased slowly over the last 5 years from 1.1 mg/dL (97 μ mol/L). Her medications are cyclosporine, azathioprine, prednisolone, simvastatin, and pancreas enzyme supplements. The main findings on examination are finger clubbing and blood pressure (BP) 148/84 mm Hg. The urine dipstick shows trace protein only. Which of the following is *true*?

 - The underlying cause of her CKD is most likely cyclosporine toxicity.
 - An urgent kidney biopsy is indicated.
 - Switching azathioprine to sirolimus is indicated.
 - If her CKD progresses, kidney transplant is contraindicated because of underlying cystic fibrosis.
 - Diltiazem should be added to control the hypertension.
- A 30-year-old man is referred to the nephrology clinic because of rising serum creatinine and worsening hypertension. Eight months ago he underwent an allogeneic stem cell transplant for acute myeloid leukemia (AML). His conditioning regimen included total body irradiation and cyclophosphamide. His early posttransplant course was relatively uncomplicated. However, he remained on cyclosporine because of GVHD of the skin. Recent cyclosporine trough concentrations have been 60 to 90 ng/mL. His medications are low doses of cyclosporine, prednisolone, valacyclovir, and sulfamethoxazole-trimethoprim. The main findings on examination are BP 166/98 mm Hg (148/86 mm Hg, 3 weeks ago), mild GVHD of the skin, and mild leg edema. The urine dipstick shows 3+ blood and 3+ protein. Other test results are as follows: white blood cells 4.2×10^9 /L, hematocrit 26.0 (32.0, 3 weeks ago), platelets 82×10^9 /L (130×10^9 /L, 3 weeks ago), creatinine 2.3 mg/dL (200 μ mol/L) (1.2 mg/dL [105 μ mol/L], 3 weeks ago), albumin 3.2 g/dL, and 24-hour urine protein 3.2 g. The prothrombin time and activated partial thromboplastin time are normal. Which of the following is *true*?

 - The main cause of the kidney disease is probably cyclosporine toxicity.
 - The patient has nephritic syndrome, most likely caused by membranous nephropathy.
 - The plasma lactate dehydrogenase, serum haptoglobin, and blood film should be checked urgently.
 - The main cause of the kidney disease is recurrence of AML, with infiltration of the kidneys.
 - Computed tomography with intravenous contrast should be performed urgently to exclude obstruction of the urinary tract.
- A 30-year-old man is referred urgently to the nephrology clinic because of an increase in creatinine from baseline 2.0 mg/dL (176 μ mol/L) to 2.9 mg/dL (255 μ mol/L). One week ago he was diagnosed with community-acquired pneumonia and treated with antibiotics. His breathlessness and cough have improved, but he reports mild headache and pain in the right big toe. Ten years ago he underwent a heart transplant for severe heart failure caused by idiopathic dilated cardiomyopathy. The heart transplant is apparently working well. His medications are cyclosporine, azathioprine, prednisolone, ramipril, atorvastatin, and (in the last week) cefuroxime plus clarithromycin. Examination shows fine hand tremor, pulse 90/min and regular, BP 142/84 mm Hg, oxygen saturation 98% on room air, normal breath sounds with no added sounds, and no leg edema. The right big toe is inflamed, consistent with acute gout. The urine dipstick shows trace protein only. Tests show serum creatinine 2.9 mg/dL (255 μ mol/L), potassium 5.9 mmol/L, white blood cells 4.9×10^9 /L (with normal differential), hematocrit 30, and platelets 189×10^9 /L. Which of the following is *true*?

 - The most likely cause of the acute kidney dysfunction is allergic interstitial nephritis resulting from cefuroxime.
 - The main underlying cause of the CKD is cardiac failure (of the transplanted organ).
 - If the patient's CKD progresses to a glomerular filtration rate below 15 mL/min, the treatment of choice would be peritoneal dialysis.
 - Allopurinol should be added to treat the gout.
 - The most likely cause of the acute kidney dysfunction is acute cyclosporine nephrotoxicity caused by impairment of cyclosporine metabolism by clarithromycin.
- A 44-year-old woman is admitted to the intensive care unit with severe, decompensated liver disease. The underlying cause is cirrhosis from hepatitis C (HCV) infection. Over 1 week she becomes oliguric; serum creatinine increases from 1.0 mg/dL (88 μ mol/L) to 2.5 mg/dL (221 μ mol/L). The urine dipstick shows trace blood and trace protein only. The fractional urine excretion of sodium is 0.2%. Low-dose noradrenaline is started, causing the mean arterial pressure to increase to 65 mm Hg and the urine output to increase to 800 mL/24 h. A decision is made to list her for an emergency liver transplant. Which of the following is *true*?

 - The baseline serum creatinine of 1.0 mg/dL likely reflects normal eGFR.
 - The clinical picture is consistent with hepatorenal syndrome.
 - The acute renal dysfunction is a result of HCV-related glomerulonephritis.
 - At this time, a simultaneous liver-kidney rather than a liver-alone transplant should be performed.
 - After successful liver transplantation, her risk for developing stage 4 or 5 CKD over the long term is negligible.
- A 39-year-old woman is referred for opinion regarding kidney transplantation. She has a background of complex congenital cyanotic heart disease and has undergone several procedures for this. However, her cardiac function is worsening; therefore, she is being considered for cardiac transplantation. Her serum creatinine has steadily increased over the last 3 years. Medications are furosemide, digoxin, bisoprolol, omeprazole. She reports poor appetite and breathlessness on minimal exertion. On examination she is thin, pulse is 114 beats/min and regular, BP 152/92 mm Hg, and respiratory rate 22/min at rest. The fingertips are cyanosed. The jugular venous pressure is increased. There are reduced breath sounds in the left lung base. The liver edge is palpable; the femoral pulses are strong with no bruits. There is trace leg edema. Tests show urine dipstick 3+ protein and trace blood, plasma creatinine 4.0 mg/dL, potassium 5.1 mmol/L, albumin 3.1 g/dL, and 24-hour urine protein 4.2 g. Which of the following is *true*?

 - The kidney dysfunction is mainly a result of the severe heart failure and thus should improve significantly after successful cardiac transplantation.
 - Because of the risk that cyclosporine or tacrolimus will worsen kidney function after cardiac transplantation, you should advise the cardiologists to avoid using these drugs after transplantation.
 - Simultaneous heart-kidney transplant is a relatively common procedure and should be considered here.
 - Simultaneous heart-kidney transplant is an uncommon procedure but should be considered here.
 - You should recommend that a kidney transplant be performed first and only later consider cardiac transplantation.

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